Quantum Mechanics Without Collapse: The Simplicity Argument

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> **Argument.** Universal unitary evolution is the dynamical baseline in this comparison. Added mechanisms must justify their explanatory cost. The comparison includes boundary data and the probability of observer records.

Abstract

Quantum mechanics can apply the same evolution rule to a particle, a measuring device and the observer. This is the starting economy of the many-worlds interpretation. Adding collapse or a separately selected particle configuration needs an explanatory benefit that justifies the extra structure. The comparison must count the complete theories, including their starting conditions and predicted observation probabilities. Many branches do not automatically mean many independent rules. Under the specified simplicity rule, a shorter complete account receives greater initial weight. The paper develops this case for Everett's interpretation and explains why the probabilities of its recorded outcomes remain essential to the comparison.

Keywords: Many-Worlds Interpretation, Simplicity Dominance Principle, Kolmogorov complexity, algorithmic probability, quantum foundations, Schrödinger equation, Born rule, measurement problem, mathematical monism

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A short law can describe a vast world. Compare the complete accounts.

1. Introduction

Why add a special change in the laws when a measurement occurs? Quantum evolution without collapse applies one rule to the measured system, the apparatus and the observer. A collapse theory changes that evolution. A Bohmian theory adds a selected configuration of particles and a rule guiding it. The question is what each complete account explains for the information it requires.

I propose an algorithmic prior for explicitly described candidates. A shorter complete description receives greater prior weight under the chosen convention. Its posterior weight also depends on its observational likelihood. Extending the prior to a measure of physical realization is a further philosophical premise.

Section 2 states the monist motivation. Section 3 distinguishes a proposed measure from the results of algorithmic information theory. The remaining

sections apply that distinction to quantum accounts and identify what must be specified before a ranking is justified.

The comparison contains both deductive and empirical questions. Given a prior proportional to 2^(−K), the weight ratio for specified descriptions follows mathematically. Given dynamics and a state, deriving predicted probabilities is likewise a mathematical task. Neither implication needs experimental confirmation to be valid. The further questions are whether the premises are justified, whether an ontological argument establishes them a priori, and which models describe the observed world. Empirically equivalent accounts cannot be distinguished by the very observations on which they agree; their assessment can still involve logical coherence and explanatory economy.

Matheism's intended contribution is an a priori account of what physical mathematics is: the structure itself, rather than a description of a second substance. That claim is different from saying that every mathematical construction belongs to our universe. Even granting the identity, a further argument must identify which structure the universe is and how representational decompositions relate to its constituents. The same quantum formalism can support competing interpretations unless additional premises select one.

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2. Reality as Mathematical Structure

2.1 The Dichotomy

All existence is either patterned or not-patterned (random). Both are mathematical.

The rule and randomness argument motivates the monist position. Algorithmic incompressibility is defined relative to an encoding, and absence of a particular pattern is not equivalent to absence of every pattern. The exhaustive partition alone does not establish the identity of reality and mathematical description.

The present comparison is conditional on explicit predictive models. It can be carried out whether or not a reader accepts the further ontological identity.

2.2 From Isomorphism to Identity

Matheism argues that causal powers and observable distinctions are structural. A proposed additional substance must explain what role it plays. This motivates identifying the physical account with its structure rather than adding an unspecified bearer.

Isomorphism alone does not imply numerical identity: distinct copies can be isomorphic. The monist identity is therefore the interpretive claim, not a consequence of using the word isomorphism. It also does not independently select a probability measure or quantum interpretation.

3. The Simplicity Dominance Principle

3.1 The Measure Problem

Universal realization of consistent structures is an additional commitment of the broader programme. Mathematical monism alone does not determine which structures observers should expect to inhabit. A description can also contain mutually incompatible requirements while remaining a consistent inscription.

A measure and an observer selection rule must be stated. Calling reality structure does not uniquely supply either.

3.2 Algorithmic Probability

I propose:

SDP: P(S) 2^(−K(S))

Here K is defined relative to a fixed reference machine and an effective description of the complete model. It is not merely the length of its displayed equation. This weighting is a proposed prior, to be combined with observational likelihoods.

Algorithmic probability has useful formal properties within its stated domain:

Dominance: Universal constructions dominate appropriate effective classes up to multiplicative constants. This is not a guarantee of best prediction on every finite dataset and does not uniquely determine an ontological measure.

Invariance: Changing a universal description language changes K by an additive constant depending on the two languages. This does not guarantee preservation of finite model rankings or of a small bit advantage.

Prediction: Simplicity can support generalization, but its contribution must be assessed together with fit to evidence. Prior preference alone cannot rescue a model with poor likelihood.

The proposed extension from inductive inference to physical realization is substantive. The mathematical results about universal priors do not prove that extension. Hutter's account of algorithmic probability states the formal setting: https://www.hutter1.net/ai/algprob.pdf.

4. Application to Quantum Mechanics

4.1 Comparing Interpretations

The comparison begins with the work each account performs. This inventory separates the shared quantum machinery from the additional task each inter-

pretation undertakes:

Account	Complete specification includes	Explanatory task
Everett	State space, Hamiltonian, boundary, subsystem structure and record weights	Explain stable records and their probabilities within unitary dynamics
Operational quantum account	Preparations, operations and measurement probabilities	Predict observations with a specified scope of application
Bohmian account	Wave dynamics, configuration, guidance and preparation measure	Account for the selected configuration and observed statistics
Objective collapse	State, modified dynamics and collapse parameters	Account for outcomes through a distinct dynamical process

Everett treats unitary evolution as complete dynamics and interprets emergent branches as realized. Those commitments avoid a dynamical collapse mechanism, but still require a definite model and an account of observed probabilities.

Measurement interactions correlate a system with an apparatus and environment. Suppression of off diagonal terms in a reduced state can make interference between records effectively inaccessible. The relevant stable states depend on the interaction and dynamics; they are not universally position eigenstates. Decoherence does not itself select a unique global outcome or establish a particular ontology (Zurek, 2003).

Objective collapse proposals such as GRW and CSL specify modified quantum dynamics; CSL uses stochastic nonlinear evolution (Bassi et al., 2013). They therefore cannot be dismissed as lacking any mathematical proposal. Their extra laws and parameters require justification, and specifying them is different from demonstrating a collapse process in nature. Everett's case is that universal unitary evolution can account for measurement without those additions. That argument must include an account of observed probabilities.

Bohmian mechanics retains a wavefunction and adds an actual configuration and a guidance law (Bohm, 1952). The parsimony objection is direct: if universal unitary evolution already accounts for records and their probabilities, what requires this extra structure? The Bohmian answer is that a definite configuration supplies a definite realized outcome. A comparison must evaluate that contribution alongside the Everettian account of records and probability. Under the proposed prior, a shorter complete account wins when likelihoods are equal. Merely counting named ingredients does not establish the shortest description.

Experimental exclusion is a different claim. In quantum equilibrium, Bohmian mechanics reproduces the usual nonrelativistic quantum statistics (Dürr, Goldstein and Zanghì, 1992). Bell violations do not exclude it through the local hidden variable bound because its guidance is nonlocal. A claimed contrary experiment must identify the particular model, preparation and differing prediction. The simplicity objection can stand without claiming such a refutation.

The present inventory does not calculate a winning complete description. Everett offers an economical candidate to assess.

One finite random measurement result has finite description length. A short stochastic law can assign a likelihood to a long record without listing that record as part of the law. Everett must likewise explain the weights of observed records. Charging exact historical selections to only one account would compare different tasks.

4.2 Bell Correlations as Structure

Bell inequalities constrain combinations of assumptions including local factorization and measurement independence. Observed violations do not refute every deterministic model or independently confirm mathematical monism.

An entangled state is a joint mathematical description of the systems. Calling it one structure does not by itself answer Bell's questions about locality and independence. Those conditions must be evaluated in the proposed model.

The observed correlations do not allow controllable signalling merely by choosing a remote measurement setting. This operational fact is distinct from whether a proposed underlying ontology is local in Bell's sense.

Matheism treats the wavefunction and its correlations as structure. Different interpretations also use mathematical descriptions, so that shared feature does not choose between them. The structural reading must recover the same observations and state what additional explanatory advantage it provides.

4.3 The Born Rule

Why do observed frequencies match $|\Psi|^2$? This remains an active research area. Deutsch (1999) and Wallace (2012) have argued that the Born rule follows from decision-theoretic rationality constraints within MWI: rational agents in branching universes must weight outcomes by squared amplitudes. Their decision-theoretic arguments use explicit assumptions about rational preferences and do not obtain weights by counting branches.

The Born rule proposals use substantive premises about rational decision, symmetries or an observer's uncertainty about location. Sebens and Carroll (2018), for example, make an explicit separability principle central to their argument. Their conclusion cannot be treated as obtained with no premises beyond the

Schrödinger equation. A simplicity comparison should state which account it adopts and include the assumptions doing the work.

Decision theoretic accounts of Born weights and algorithmic model comparison address different questions. A successful comparison must include how each interpretation assigns probabilities to observations. A stochastic collapse law need not encode every outcome as part of its law, and an Everettian law does not remove the need for observational weights. This paper has not selected an interpretation merely by assigning infinite outcome cost to its rival.

A short code assigned using known Born weights would encode those weights, not independently derive them. No general theorem here identifies a branch's Kolmogorov complexity with the negative logarithm of its squared amplitude. That proposed connection remains a separate research question.

4.3.1 A refinement test for record weights

Consider orthogonal records |0> and |1> correlated with orthogonal environmental states, with global state

$$|\Psi\rangle = \frac{\sqrt{3}}{2}|0, E_0\rangle + \frac{1}{2}|1, E_1\rangle.$$

The squared norms of the two record components are 3/4 and 1/4. Now refine only the environment correlated with record 0 by an isometry sending E_0 to an equal superposition of m orthogonal substates E_0j, while E_1 remains orthogonal to them. Such an isometry can be extended to a unitary on a sufficiently large environment. The resulting expression is

$$|\Psi_m\rangle = \sum_{j=1}^{m} \frac{\sqrt{3}}{2\sqrt{m}}|0, E_{0j}\rangle + \frac{1}{2}|1, E_1\rangle.$$

Tracing out the environment gives the same record state, diag(3/4,1/4), for every positive integer m. Counting the displayed terms instead would assign record 0 the fraction m/(m+1), changing from 1/2 at m=1 to 3/4 at m=3 and 9/10 at m=9. The observed record distinction has not changed.

This calculation gives an exact consistency requirement: an account that assigns record weights from the reduced record state must be invariant under this environmental refinement. Squared-norm weights satisfy it; naive equal counting of displayed components does not. Treating reduced-state invariance as a requirement on rational expectation is an additional premise. The example does not derive the Born rule from unitary evolution alone. It identifies why an invariant weighting rule, rather than a count of written branches, belongs in the complete comparison. Wallace's decision-theoretic programme makes its rationality assumptions explicit (Wallace, 2009).

4.4 Clarification

The universal realization premise permits considering different consistent dynamics. It does not say that every informal interpretation is a complete model or that all such models recover our observations.

A stochastic model can have a short law even when a typical individual outcome sequence is incompressible. Its outcome probabilities enter the likelihood. A deterministic model must likewise account for initial conditions and observed records, and an Everettian model must connect branch weights to observations. Charging random outcomes only to collapse models while charging no observational selection to rivals does not provide a fair comparison.

No ranking of the complete candidate models has been established here.

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5. Objections

"MWI adds infinite branches-that's complex!"

Branches are consequences, not axioms. Complexity is measured by description length, not by what the description entails. A short program can generate infinite output. The Schrödinger equation is simple; its solutions are vast.

"This is untestable metaphysics."

A specified dynamical collapse model can make different predictions from unitary quantum mechanics. Those predictions can be tested. Quantum computation by itself does not select Everett over other accounts reproducing the same quantum statistics. Likewise, a simplicity prior need not assign zero probability to a complex model; a difficult outcome is not automatically a logical falsification of the prior.

"Why algorithmic probability?"

Universal induction has precise dominance properties over specified computable model classes. The choice of reference machine, model class and physical observation rule still matters. These results do not establish a unique ontological measure.

"The ontological question is misguided."

Pragmatist and Bayesian interpretations pose a different question about the role of quantum states. The present proposal compares explicitly predictive models rather than claiming that a measure argument has already refuted those interpretations.

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6. The Three-Way K-Comparison

Use one explanatory task and encoding. For predictive comparison, the score includes a model description and the information cost of its data under its likelihood. A joint description can exploit shared information; K(L,I,O) is not generally the sum of three unconditional complexities.

Collapse: Include the actual stochastic dynamics and its parameters. Evaluate observed outcomes through its likelihood. A finite observed bit string is not an infinite specification of the law.

Everett: Include the quantum dynamics, boundary data and the account assigning weights to observers' records. Saying that all outcomes occur does not make every observed record equally probable or remove the need to explain observed frequencies.

Superdeterminism: Include the common generator, preparation measure and response rules. Measurement dependence can have a short description while still failing empirical or robustness tests.

Two additional points need explicit treatment in an Everettian comparison:

The Born rule. State the assumptions used to obtain observational weights and include them in the account. Its cost cannot be declared zero simply because an argument is labelled a derivation.

The stable record basis. Specify the interactions, subsystem decomposition and initial data underlying decoherence. Reusable structure may compress these commitments, but they cannot be omitted from one candidate and counted against another.

The comparison is conditional on complete models, the coding convention and the evidence. If their likelihoods agree, a justified shorter description retains its prior advantage. If the likelihoods differ, those differences also matter.

7. Exact histories and predictive laws

A demand for causal completeness can motivate rejecting irreducible outcome selection. The strongest form of that objection asks for a generator of the particular history, rather than stopping at its probability distribution. The mathematical consistency of a stochastic law and the explanatory sufficiency of irreducible chance are separate questions.

An insistence that every realized history must have a finite deterministic generator is a further constructive premise. Probability theory can consistently describe measures whose typical infinite realizations lack such generators. Excluding those realizations by premise is not a proof that stochastic laws are contradictory. Deterministic theories must also constrain their initial states if every history is to be computable. The preference for a fully constructive account belongs in the research programme, with its assumptions explicit.

8. Records within a unitary substrate

A unitary evolution can support records whose local description is effectively lossy because information remains in environmental correlations. A many to one map of local records has no unique inverse on its full input domain. Reversible computation can retain inputs in an enlarged state; the lack of a local inverse does not prohibit reversed physical solutions or their observers.

This compatibility removes the need to introduce collapse solely to account for inaccessible local information. It does not independently derive the thermodynamic arrow or decide between complete quantum interpretations. Boundary conditions, record reliability, outcome probabilities and observational predictions remain part of the comparison. SDP offers a proposed prior for complete descriptions; it has not established that low entropy boundaries dominate that prior.

Observer development within the quantum account

Simple laws can remain unchanged while the structures they generate acquire the organization required for observers. Suppose the starting state lacks that organization and a later state contains observers who require it. The history must then develop the missing organization. This is a necessary consequence of those premises, not merely a possible outcome. The growing quantity is the observer relevant organization of the generated states; the law and the shortest description of the history can remain simple.

The requirement applies to any interpretation offered as an account of our developmental history. It does not by itself select MWI over another interpretation that recovers that history.

9. Conclusion

The economy of quantum mechanics without collapse is that one evolution rule covers systems, apparatus and observers. Many outcomes within its solution do not require a separate law for each outcome.

The simplicity argument becomes decisive between specified candidates only when their complete descriptions and predicted probabilities are compared. Starting conditions count. The explanation of recorded outcomes counts. Under the declared weighting rule, a shorter account with the same fit to observations has the corresponding advantage. That is the case for Everett developed here and the standard its competitors can answer.

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